

CSC-335

HCI & Computer Graphics

Lec 3: User-Centered Design

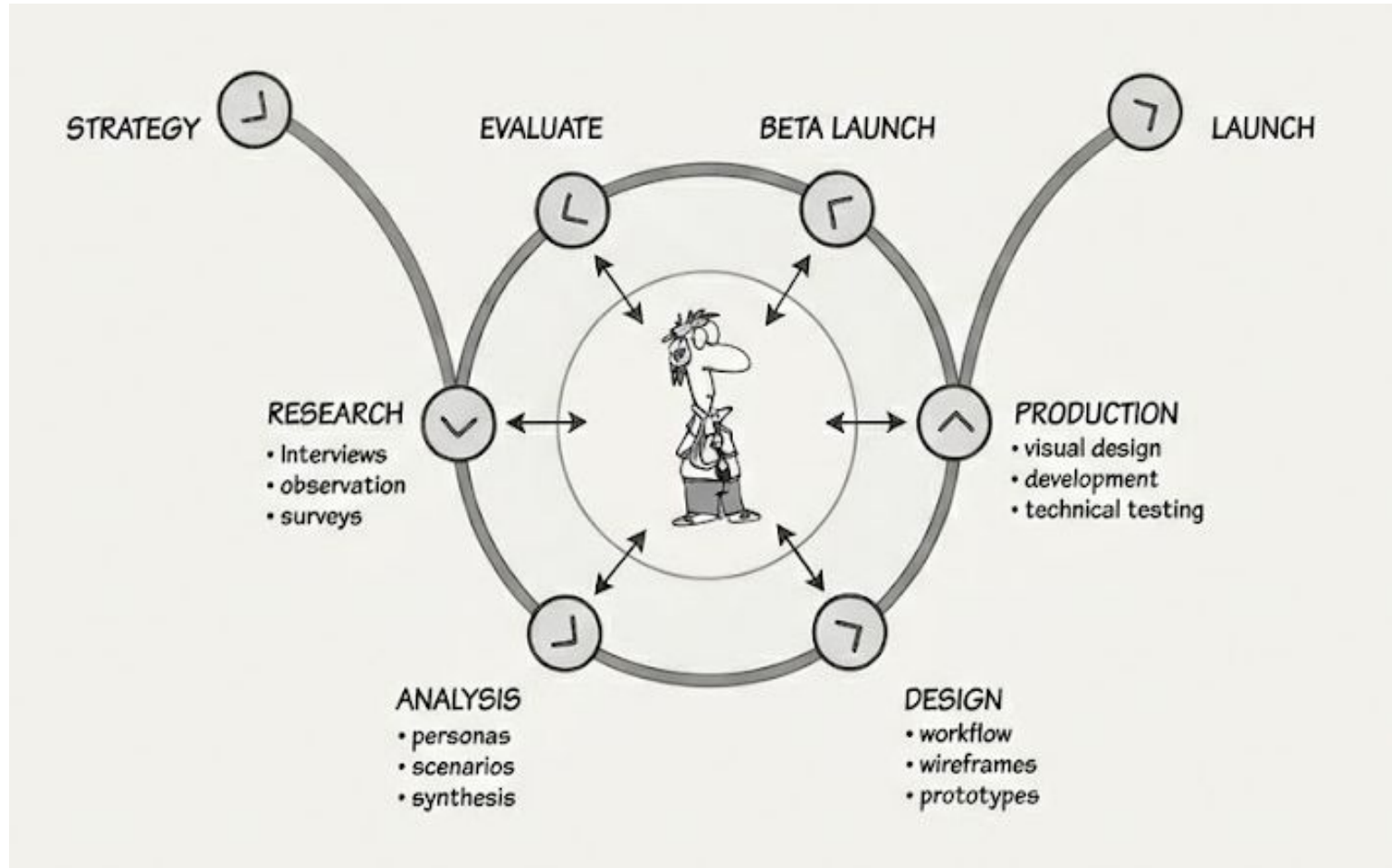


The Essence of User-Centered Design

User-Centered Design (UCD) is an iterative design process in which designers focus on the users and their needs in each phase of the design process. UCD calls for involving users throughout the design process via a variety of research and design techniques so as to create highly usable and accessible products for them.

UX Process

UX Process Overview



UX Process - Phase 1: Strategy & Research

Strategy: Setting the Course

- Articulates the brand's guiding principles and long-term vision.
- Shapes project goals: what to achieve, how to measure success, and its priority.
- Defines the foundational purpose and direction for the entire project.

Research: Uncovering Insights

Also known as the Discovery phase, this involves:

- Gathering user and competitor data through various methods (interviews, surveys, observations).
- Crucial for creating an informed user experience.
- Often undervalued or skipped in "Lean UX" approaches, despite its importance.



UX Process - Phase 2: Analysis & Design

Analysis: Interpreting Data

- Draws actionable insights from data collected during the research phase.
- Organizes and infers meaning from observations to understand the "why."
- Communicating findings back to end-users validates assumptions.

Design: Crafting Solutions

- A collaborative and iterative process.
- Involves putting ideas in front of users, gathering feedback, and refining concepts.
- Utilizes low-fidelity prototypes (paper, wireframes) to focus on functionality over aesthetics.



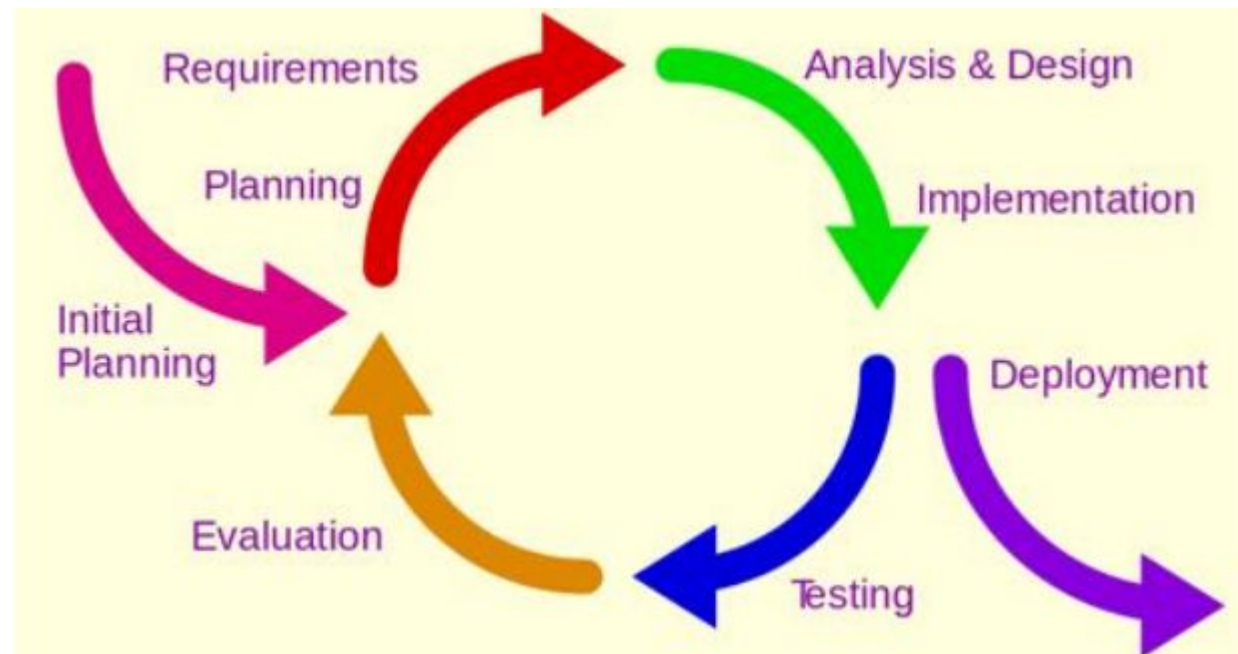


UX Process - Phase 3: Production

The production phase brings the design to life with meticulous attention to detail:

- High-fidelity designs are fully fleshed out, incorporating all visual elements and branding.
- Content is finalized and digital assets are created or integrated.
- The near-final product undergoes rigorous user testing and validation with stakeholders and end-users to ensure it meets all requirements and delivers an optimal experience.

Rapid Prototyping or Iterative Design

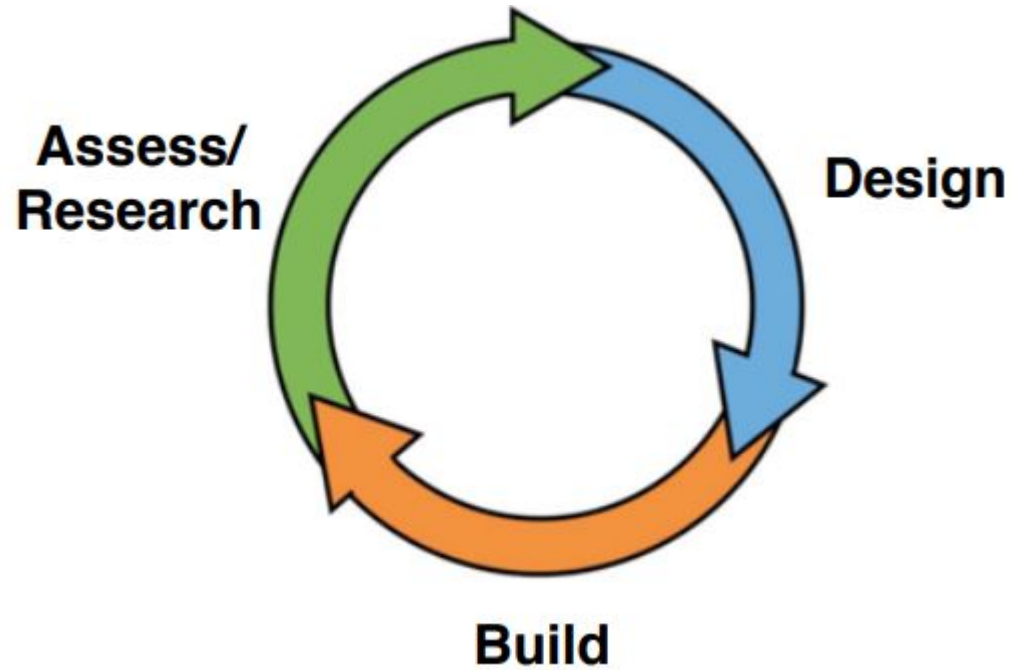


- <https://www.interaction-design.org/literature/article/design-iteration-brings-powerful-results-so-do-it-again-designer>

LeanUX



Iterative Design



Iterative Design - 1

- Assess

- what users are currently doing
- what their needs are
- understand the design space
- the problems that they're facing

- Design

- we take what we've learned in the initial assessment
- come up with ideas, lots of ideas

Iterative Design - 2

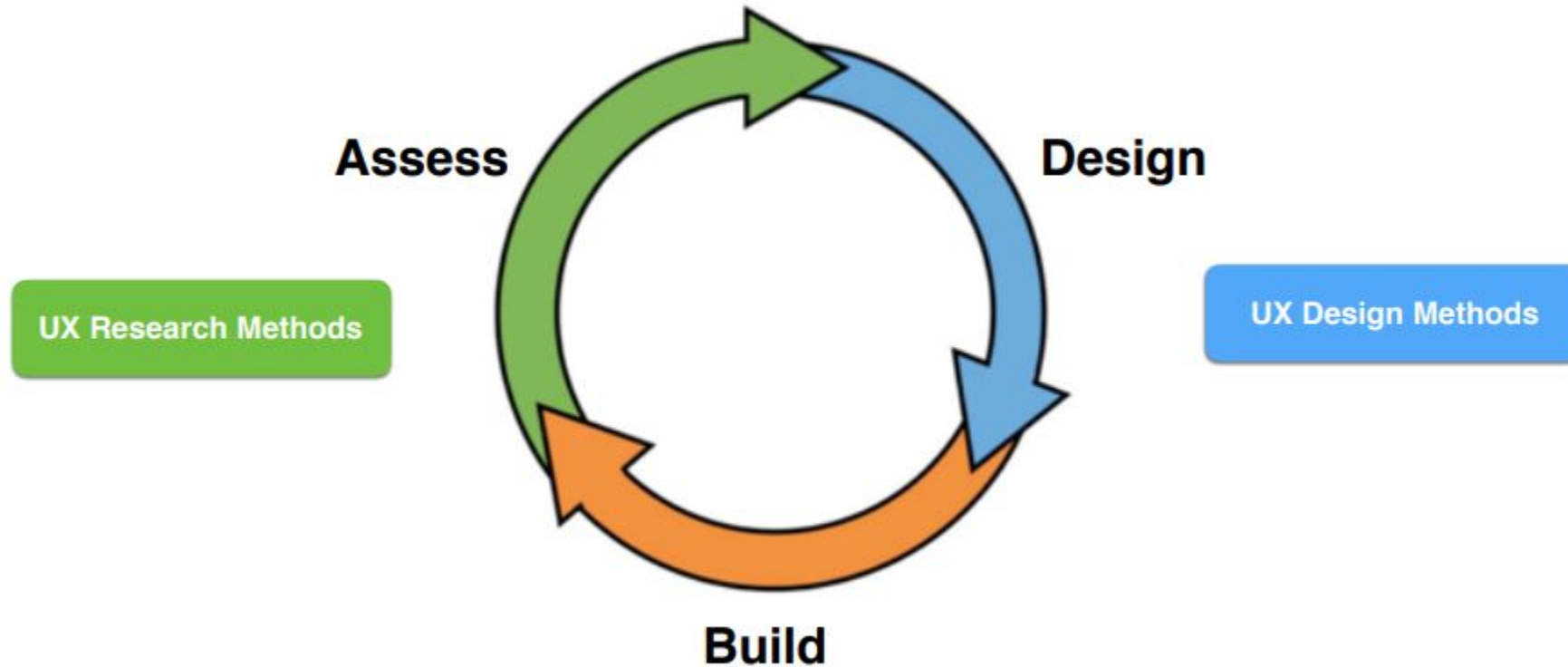
- Build

- we take those design ideas or a subset
- we build them into prototypes or some kind of representation that we can use to communicate a particular idea
- get feedback on whether that idea is headed in the right direction

- (Re)-Assess

- Re-Apply assessment methods, like user testing or formal inspection methods, to see if that prototype is leading towards a good user experience or a bad user experience
- We take what we learn from that, and we repeat it all again
- We go back to the drawing board
- We design new ideas
- We build new prototypes
- We perform new assessments
- And we continue this until we're satisfied that the product is ready to go.

Integrate Research and Design



Iterative Design - 2

- UX Research: Key Methods

- Interviews
- Observations
- Surveys
- User Testing
- Inspection Methods

- UX Design : Key Methods

- Personas, Scenarios, User Stories
- Sketching and Ideation
- Storyboarding
- Mapping and Navigation Design
- Comparative Research
- Lo-, Mid-, & Hi-Fidelity Prototyping